

2 (**long**) Consider the incompressible, steady flow in a pipe with an orifice plate as shown in Fig. 2.

- (a) Find the difference in static pressure, $(p_1 - p_2)$, between the entrance and the orifice in terms of the inlet dynamic head and A_1 and A_2 . State clearly your assumptions. [10]
- (b) Find the difference in static pressure, $(p_2 - p_3)$, between the orifice and the pipe exit in terms of the inlet dynamic head and A_2 and A_3 . Again, state clearly your assumptions. [10]
- (c) By considering an appropriate control volume, find the force on the orifice plate in terms of the inlet dynamic head and the areas A_1 and A_2 . Again, state clearly and justify any assumptions you make. [10]

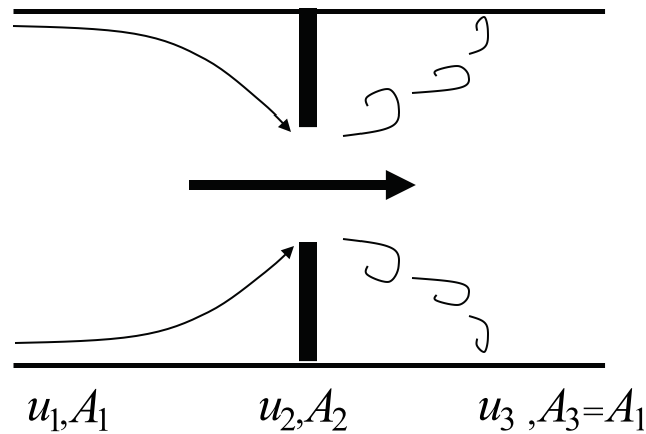


Fig. 2